

Act V: Different Stages

Market Structures

Module XVI: The Boardroom

Oligopoly and Game Theory



Previously in Marginal Falls

- ◆ Farmers Market → many sellers
- ◆ Corporate District → one seller
- ◆ Main Street → differentiated sellers

Next: Lantern Island is different.

A New Kind of Market

In some markets, there are only a few firms.
Each firm is large enough to affect the market.
Each firm must watch what the others do.

Oligopoly:

A market structure with a small number of interdependent firms.

Why Oligopoly is Different?

Perfect competition: One seller is too small to matter.

Monopoly: One seller has no rival.

Oligopoly: Each firm has rivals, and those rivals matter.

Oligopoly = market power + strategic interaction

Oligopolies exist because entry is difficult.

Common barriers:

- ◆ economies of scale
- ◆ control of key inputs
- ◆ government licenses
- ◆ patents
- ◆ large startup costs

Economies of Scale

Some industries require large fixed costs before production begins.

Once the system exists, serving more customers becomes cheaper per customer.

Large firms can spread fixed costs over many buyers.

Small firms cannot.



Other Barriers to Entry

Entry can also be blocked by:

Key input: one or few firms control what is needed.

License: only approved firms can operate.

Patent: legal protection prevents copying.



Choose a price

Econgames.promitkchaudhuri.com/duopoly

Choose one fare:

Premium Fare: \$18

Discount Fare: \$14

Your rival chooses at the same time.

Your profit depends on both choices.

Game Theory

In monopoly, River-Gate asked:
“How many passes should I sell?”

In oligopoly, each firm asks:
“What should I do if my rival changes price?”

The firm's profit depends on the rival's action.

Game theory studies situations where one player's payoff depends on the choices of other players.

What is a Game?

A game has:

Players: who makes decisions.

Strategies: what each player can choose.

Payoffs: what each player gets from every combination of choices.

For oligopoly:

Players = firms

Strategies = prices, output, advertising, quality

Payoffs = profits

The Payoff Matrix

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	(750, 250)	(350, 350)

A payoff matrix shows what each firm earns from every combination of choices.

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	(750, 250)	(350, 350)

Suppose Ridgeway chooses Premium.

Copperline earns:

Premium: 500

Discount: 750

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	(750, 250)	(350, 350)

Suppose Ridgeway chooses Premium.

Copperline earns:

Premium: 500

Discount: 750

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	 (750, 250)	(350, 350)

Suppose Ridgeway chooses Premium.

Copperline earns:

Premium: 500

Discount: 750

So Copperline chooses ***Discount***.

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	 (750, 250)	(350, 350)

Suppose Ridgeway chooses Discount.

Copperline earns:

Premium: 250

Discount: 350

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	 (750, 250)	(350, 350)

Suppose Ridgeway chooses Discount.

Copperline earns:

Premium: 250

Discount: 350

Copperline's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	(750, 250)	(350, 350)

Suppose Ridgeway chooses Discount.

Copperline earns:

Premium: 250

Discount: 350

So Copperline chooses ***Discount***.

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	★ (750, 250)	★ (350, 350)

Suppose Copperline chooses Premium.

Ridgeway earns:

Premium: 500

Discount: 750

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750)
Copperline Discount	★ (750, 250)	★ (350, 350)

Suppose Copperline chooses Premium.

Ridgeway earns:

Premium: 500

Discount: 750

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) ★
Copperline Discount	★ (750, 250)	★ (350, 350)

Suppose Copperline chooses Premium.

Ridgeway earns:

Premium: 500

Discount: 750

So Ridgeway chooses ***Discount***.

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) ★
Copperline Discount	(750, 250) ★	(350, 350) ★

Suppose Copperline chooses Discount.

Ridgeway earns:

Premium: 250

Discount: 350

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) ★
Copperline Discount	★ (750, 250)	★ (350, 350)

Suppose Copperline chooses Discount.

Ridgeway earns:

Premium: 250

Discount: 350

Ridgeway's Best Responses

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) 
Copperline Discount	(750, 250) 	(350, 350)  

Suppose Copperline chooses Discount.

Ridgeway earns:

Premium: 250

Discount: 350

So Ridgeway chooses ***Discount***.

Dominant Strategy

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) ★
Copperline Discount	★ (750, 250)	★ (350, 350) ★

A dominant strategy is the best choice no matter what the rival does.

For **Copperline**:

Discount is the dominant strategy.

For **Ridgeway**:

Discount is the dominant strategy.

Nash Equilibrium

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) ★
Copperline Discount	★ (750, 250)	★ (350, 350) ★

A Nash equilibrium occurs when each player chooses the best response to the other player.

Here:

Copperline chooses Discount.

Ridgeway chooses Discount.

Nash equilibrium:

(Discount, Discount) = (350, 350)

The Oligopoly Puzzle

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	 (500, 500)	 (250, 750)
Copperline Discount	(750, 250)	(350, 350)

Both firms would prefer:
(Premium, Premium) = (500, 500)

But each firm has an incentive to cut price.

So the market moves to:
(Discount, Discount) = (350, 350)

Collusion

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	 (500, 500)	 (250, 750)
Copperline Discount	(750, 250)	(350, 350)

Collusion is an agreement among firms not to compete.

If Copperline and Ridgeway agree to keep Premium fares, both earn more.

(Premium, Premium) = (500, 500)

(Discount, Discount) = (350, 350)

But each firm is tempted to secretly cut price.

Cooperative vs Noncooperative Outcomes

Outcome	What Firms Do	Result
Cooperative outcome	Coordinate	(Premium, Premium)
Noncooperative equilibrium	Act independently	(Discount, Discount)

Prisoner's Dilemma

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) 
Copperline Discount	(750, 250) 	(350, 350)  

This game is a prisoner's dilemma.

Each firm follows its dominant strategy.

But when both follow their dominant strategy, both are worse off than if they had cooperated.

Prisoner's Dilemma

	Ridgeway Premium	Ridgeway Discount
Copperline Premium	(500, 500)	(250, 750) 
Copperline Discount	(750, 250) 	(350, 350)  

Dominant strategy: Discount

Nash equilibrium: Discount,
Discount

Jointly better outcome: Premium,
Premium